URINALYSIS AI

Clinical & Scientific Report

Smartphone-Based Urinalysis for Pregnancy Detection:
Methodology, Performance, Clinical Review & Phase 2 Protocol

Version 2.0 | May 2026

Research Use Only - Not a Diagnostic Device

Reviewed by independent AI clinical panels (Perplexity, Gemini, ChatGPT)

101 subjects | 1,222 images | 68 video clips | GroupKFold cross-validation

1. Executive Summary

Urinalysis AI is a research-stage system that analyzes smartphone-captured images and videos of urine samples to estimate pregnancy probability. The system uses machine learning models trained on visual features (color, texture, turbidity, spectral ratios, temporal dynamics) extracted from standardized urine photographs and short video clips.

Current status: The system has been evaluated on 101 subjects (46 pregnant, 55 non-pregnant) using rigorous GroupKFold cross-validation with complete subject-level separation. After discovering and correcting multiple sources of data leakage in earlier model versions, the honest performance metrics are:

Model	Subjects	AUC	Sensitivity	Specificity	Notes
V6 Production (urine only)	101	0.749	0.602	0.734	Pure visual features, rich aggregation
V6 Fusion (photo+video)	68	0.770	0.694	0.713	Late fusion 0.7P+0.3V, lowest variance

Key finding: The model detects a signal above chance (AUC 0.50), but the signal is weak and far below clinical diagnostic thresholds (AUC > 0.95). Three independent expert clinical reviews (Perplexity, Gemini, ChatGPT/GPT-5) unanimously conclude that: (a) a biological signal likely exists but is faint; (b) the most plausible shortcut the model may exploit is riboflavin fluorescence from prenatal vitamins, not pregnancy-specific urine changes; (c) the system cannot replace standard pregnancy tests in its current form; and (d) a rigorous Phase 2 protocol is needed to determine whether the signal is real and clinically useful.

This report presents all findings with full transparency, including failed experiments and leakage discoveries, to enable informed clinical evaluation.

2. Scientific Background & Biological Plausibility

2.1 The Central Question

Can pregnancy be detected from the macroscopic visual appearance of urine - color, clarity, texture, and temporal behavior - as captured by a smartphone camera, without chemical reagents or test strips?

2.2 What Pregnancy Does to Urine

During pregnancy, several physiological changes affect urine composition and appearance:

- **Glomerular Filtration Rate (GFR):** Increases by approximately 50% due to progesterone and relaxin, leading to diluted urine with lower urobilin concentration. This manifests as lighter, more transparent urine in many pregnant women.
- **hCG (Human Chorionic Gonadotropin):** The primary pregnancy biomarker. However, hCG itself is optically invisible at physiological concentrations. It does not change urine color, turbidity, or any visually measurable property. Standard pregnancy tests use antibody-based immunoassays to detect it - a chemical, not optical, process.
- **Protein excretion:** Mildly elevated in normal pregnancy, potentially affecting surface tension (foam patterns) and light scattering (turbidity). In pathological states like preeclampsia, proteinuria becomes more pronounced.
- **Hormonal metabolites:** Estrogen metabolites (estriol-3-glucuronide), DHEA-S, progesterone metabolites, and carnitine derivatives change throughout pregnancy. Metabolomics studies have shown these can predict gestational age (R=0.95 in development, R=0.79 in external validation) - but through mass spectrometry, not visible appearance.
- **Prenatal supplements:** Folic acid (vitamin B9) and riboflavin (vitamin B2) in prenatal vitamins are excreted in urine and produce bright fluorescent yellow coloration. This is the single most likely confounding optical signal.

2.3 Biological Plausibility Assessment

All three independent clinical reviews converge on the same assessment:

2.3.1 Signal Ranking (Strongest to Weakest)

The most plausible real visual signals, ranked by likelihood of detection via smartphone camera:

1. **1. Concentration and hydration proxies:** Color intensity reflecting urobilin concentration, strongly influenced by GFR changes. AUC for dehydration detection from urine photos has been demonstrated at clinically useful levels (Bustam et al., 2023). However, this is a hydration signal, not a pregnancy-specific signal.
2. **2. Surface tension and foam dynamics:** Protein-induced changes in surface tension could theoretically alter foam formation and decay patterns visible in video. This is speculative but physically plausible.
3. **3. Turbidity and light scattering:** Changes in dissolved solutes, crystalluria, or cellular elements (leukocytes in UTI, common in pregnancy) can alter light scattering. CIE Lab* analysis shows strong correlation between turbidity and L* value (AUC=0.984 for clear vs. turbid classification).

4. **4. Refractive index shifts:** Large glycoproteins in urine could subtly affect light refraction patterns visible as specular highlights in photos. Extremely weak signal.
5. **5. Direct hCG optical effect:** None. hCG is optically invisible at pregnancy concentrations. This is not a viable detection pathway.

2.3.2 The Riboflavin Hypothesis

The most critical confounder identified across all reviews is **riboflavin (vitamin B2) from prenatal supplements**. In our dataset, 45% of pregnant subjects report taking folic acid vs. only 7% of non-pregnant subjects. Prenatal vitamins typically contain riboflavin, which produces a distinctive fluorescent yellow color in urine. The model may be detecting supplement use - a behavioral marker of pregnancy awareness - rather than any physiological pregnancy signal.

The decisive experiment: Train three separate models: (1) folic-acid-only logistic regression; (2) urine-visual-features-only; (3) combined. If model (1) alone achieves high AUC (>0.75), and model (3) improves substantially over (2), then almost all of the gain is behavioral, not biological. This experiment has not yet been conducted.

2.3.3 Literature Context

Key findings from the scientific literature that inform this assessment:

- **Bustam et al., 2023 (Digital Health):** Smartphone urine photo colorimetry can predict dehydration with good accuracy using a controlled photo box. Performance depends strongly on illumination and device conditions.
- **Prabhakar et al., 2022 (Scientific Reports):** Even with direct metabolomics access (1H-NMR), urinary metabolite signatures for early pregnancy detection in buffaloes achieve AUC around 0.80 - making a 0.95 AUC target from bulk RGB appearance biologically ambitious.
- **Flaucher et al., 2022 (IEEE JTHM):** Smartphone urinalysis succeeds when there is a reagent strip that creates a clear colorimetric target. Raw urine appearance alone has not been shown to support comparable accuracy.
- **Shen et al., 2025 (Briefings in Bioinformatics):** LC-MS/MS metabolomics on 346 samples shows urine carries rich gestational age information - but through molecular assays, not macroscopic visual appearance.
- **Noor Azhar et al., 2023:** Color card calibration significantly improves inter-phone and intra-phone agreement for urine colorimetry, reinforcing that our current protocol needs standardization.

3. Methodology

3.1 Data Collection

Subjects provided urine samples photographed and/or videographed using their personal smartphones under semi-controlled conditions:

- **Protocol:** White background surface, approximately 30 cm distance, natural daylight, transparent cup. Three still photos and one 10-second video per subject (when available).
- **Subjects:** 101 total (46 confirmed pregnant via blood/urine test, 55 confirmed non-pregnant). One ambiguous subject (N57) excluded due to contradictory survey responses.
- **Images:** 1,222 photographs across all subjects.
- **Videos:** 68 video clips from subjects with video-capable submissions.
- **Survey data:** Age, gestational age, first morning urine, medications/supplements, health conditions, phone model, coffee and fluid intake.

3.2 Feature Extraction

221 features per subject computed from photos and videos:

3.2.1 Photo Features (per image, then aggregated per subject)

- Color features: RGB means, HSV means, color variance, hue distribution
- Texture features: edge intensity, Laplacian variance, local binary patterns
- Clarity features: turbidity index, brightness, contrast
- Spectral ratios: R/G, R/B, B/G ratios, Layla ratios

3.2.2 Aggregation Strategy (V6 Innovation)

Unlike earlier models that used only mean values across images, V6 computes rich per-subject statistics:

- Mean, standard deviation, minimum, maximum, coefficient of variation, and range for each base feature
- Interaction features: cross-channel products (e.g., R_mean_std x B_mean_std)

Key finding: Variability features (std, CV, range) consistently outperform absolute values as discriminators. The top feature is edge_intensity_max (importance=0.095), followed by video temporal complexity (0.090). This pattern suggests the model may be detecting capture instability rather than stable biophysical differences - a concern flagged by all three clinical reviewers.

3.2.3 Video Features

- Temporal statistics: frame-to-frame changes in color, texture, brightness
- Autocorrelation features: temporal stability of visual properties
- 44 temporal features per video clip

3.3 Model Architecture

V6 Production: Gradient Boosting Machine (GBM) with mutual information feature selection (k=50-90), depth=2, learning rate=0.05.

V6 Fusion: Late fusion combining 0.7 * photo-only model + 0.3 * video-only model predictions. Requires both photo and video data (available for 68 subjects).

3.4 Evaluation Protocol

- **Cross-validation:** 5-fold GroupKFold with complete subject-level separation. All images and videos from a single subject are always in the same fold.
- **Feature selection:** Mutual information (SelectKBest) performed inside each fold, not globally. This prevents information leakage from test subjects into feature selection.
- **No survey features:** All survey-derived variables (age, coffee, first_morning, folic acid, medications, gestational age) are excluded from model input to prevent behavioral and label leakage.
- **Metric:** Primary metric is AUC-ROC. Sensitivity and specificity reported at Youden's optimal threshold.

4. Data Integrity: Leakage Discovery & Correction

A critical part of this project's evolution has been the **discovery and correction of multiple forms of data leakage**. We report these transparently as evidence of methodological rigor.

4.1 Label Leakage via gest_age_weeks

The feature gest_age_weeks (gestational age in weeks) is defined as 0 for all non-pregnant subjects and >0 for all pregnant subjects. Including it in a binary pregnancy classifier is equivalent to partially re-using the label as an input. A single-feature classifier using only gest_age_weeks achieves AUC approaching 1.0.

Impact: Inflated V5 model AUC from 0.774 to 0.907. This number is invalid and should be disregarded entirely.

Correction: gest_age_weeks permanently excluded from all production models.

4.2 Behavioral Leakage via has_folic_acid

Folic acid supplementation is a strong behavioral proxy for pregnancy awareness. In our dataset, 45% of positive subjects report folic acid use vs. 7% of negatives. Including this feature allows the model to detect supplement-taking behavior, not urine physiology.

Impact: Contributed approximately 0.03 AUC inflation when included with other survey features.

Correction: Excluded from all production models. Can only be reintroduced after recruiting a balanced cohort where non-pregnant TTC women also take folic acid.

4.3 Survey Format Artifact

Old survey format (61 subjects: P1-P34/P41, N1-N28) did not capture age, coffee, or first_morning, defaulting to 0. New format (40 subjects: P42-P52, N31-N60) captured these fields. Since old format was enriched in positive subjects (21/46 positive have age=0 vs. 3/55 negative), any model including these fields learned 'old format likely positive' rather than genuine biological patterns.

Correction: All survey features excluded until old-format subjects can be re-surveyed with complete data.

4.4 Remaining Leakage Risks

Identified but not yet fully mitigated:

- **Model selection on same CV folds:** Multiple model configurations were tested on the same 5 folds. The reported 'best' AUC may be slightly optimistic (estimated +0.02-0.05). Mitigation: nested CV required for Phase 2.
- **Phone/lighting confounding:** Different recruitment waves used different phone models and lighting conditions. If pregnancy prevalence correlates with device type, the model may learn device signatures. Mitigation: standardized imaging protocol required for Phase 2.
- **Variability features as scene physics:** The dominance of std/CV/range features in model importance may indicate the model is reading scene-cup-liquid system instability (meniscus, bubbles, reflections, auto-exposure) rather than biology.

4.5 Model Performance History

Complete performance timeline with leakage status:

Model	AUC	Status	Issue
V1 Video RF (Apr 30)	0.825	INVALID	Overfit, low specificity (0.480)
V3 Ensemble (May 3)	0.854	INVALID	Non-reproducible fold split
V4 Urine only (May 4)	0.739	VALID	Pure urine, mean aggregation
V5 + survey (May 4)	0.774	PARTIAL	Survey format artifact (+0.03)
V5 Leaky (May 4)	0.907	INVALID	Label leakage (gest_age_weeks)
V6 Production (May 4)	0.749	VALID	Honest, rich aggregation, no survey
V6 Fusion (May 4)	0.770	VALID	Best model, photo+video, std=0.043

5. Confounder Analysis

5.1 Known Confounders

Confounder	Mechanism	Severity	Current Mitigation
Hydration status	GFR changes dilute urine, changing color/clarity independently of pregnancy	HIGH	First-morning flag collected but not controlled
Prenatal vitamins (riboflavin)	Bright fluorescent yellow urine from B-vitamin excretion	HIGH	Excluded from model; ablation study needed
Phone model	Different cameras produce different color renderings	MEDIUM	Recorded but not normalized
Lighting conditions	Natural daylight varies significantly	MEDIUM	White background used; lightbox needed
UTI	Turbidity, color changes from infection (17% of positives have UTI history)	MEDIUM	Recorded as health condition
Coffee/caffeine	Diuretic effect changes concentration	LOW	Recorded in survey
Time of day	Urine concentration varies throughout day	LOW	First-morning flag collected
Medications	Various drugs can alter urine color	LOW	Recorded in survey

5.2 The Riboflavin Ablation (Not Yet Conducted)

All three clinical reviews identify this as the single most important experiment:

- **Design:** Recruit a matched cohort where non-pregnant women trying to conceive also take prenatal vitamins (including folic acid and riboflavin). Compare model performance on this 'riboflavin-balanced' cohort vs. the current unbalanced cohort.
- **Expected outcome under H0 (no real signal):** AUC drops to chance (~0.55) when riboflavin is balanced between groups.
- **Expected outcome under H1 (real signal exists):** AUC remains above 0.65 even with riboflavin balanced.
- **No-go rule:** If AUC drops to chance, the project should terminate as a pregnancy detector and consider pivoting to other urinalysis applications.

5.3 Demographic Profile

Characteristic	Positive (n=46)	Negative (n=55)
Age range	N/A-36 (many missing)	19-36
Mean age (where available)	N/A	23.1
Folic acid use	45% (of surveyed)	7%
Coffee drinkers	N/A	50%
First morning urine	N/A	17%
Health conditions	0% (of surveyed)	17%
Has medications	55% (of surveyed)	37%

6. Independent Clinical Review Panel

Three independent AI clinical review systems were provided with complete methodology documentation and asked to evaluate: (a) methodology validity; (b) effort allocation for improvement; (c) falsification experiment design; (d) feature engineering priorities; and (e) validation requirements. A fourth review (ChatGPT) provided a comprehensive Hebrew-language clinical literature review. Key findings are synthesized below.

6.1 Consensus Findings (All Reviewers Agree)

- **0.774/0.749 is the honest number.** 0.907 is invalid due to label leakage. All reviewers confirm this assessment independently.
- **hCG is optically invisible.** No reviewer believes hCG can be detected from urine appearance at physiological concentrations.
- **Riboflavin is the most likely shortcut.** Prenatal vitamin excretion, not pregnancy physiology, is the most probable source of the detected signal.
- **Variability features are suspicious.** The dominance of std/CV features suggests scene physics (meniscus, bubbles, auto-exposure) rather than biology.
- **Current performance is not clinically useful.** AUC 0.75 means ~25% error rate, far below the >99% sensitivity required for pregnancy screening.
- **More data and better protocol are the primary levers.** Not more sophisticated modeling.
- **Phase 2 must include serum beta-hCG ground truth.** Self-report and home urine tests are insufficient for validation.

6.2 Effort Allocation (Reviewer Consensus)

Lever	Perplexity	ChatGPT	Average	Rationale
A. Data Scale	35%	45%	40%	N=101 is too small; CI width ~0.08-0.10
B. Protocol	30%	35%	32.5%	Illumination control is critical for colorimetry
C. Features	25%	15%	20%	DINOv2/VideoMAE may help after N increases
D. Modeling	10%	5%	7.5%	Model shopping is mostly optimism shopping at this N

6.3 Falsification Experiment Design

Both structured reviewers (Perplexity and ChatGPT) converge on a similar kill-study design:

- **Sample size:** 120-240 new subjects (60-120 per class), matched on hydration, time-of-day, phone model, and critically, supplement use.

- **Ground truth:** Serum beta-hCG (quantitative), not self-report or home test.
- **Protocol:** Fixed lightbox, color calibration card, standardized cup, fixed distance and angle.
- **Pipeline:** Pre-registered and frozen before data collection. No post-hoc tuning.
- **Primary metric:** Subject-level AUROC with 95% bootstrap CI and permutation test (H0: AUROC <= 0.60).
- **No-go rule:** If locked model AUROC < 0.70 with lower CI including 0.60, or if metadata-only model achieves AUROC > 0.60, the project should terminate as a pregnancy detector.

6.4 Pivot Opportunity: Preeclampsia/Proteinuria

The Gemini review identifies a significant pivot opportunity. Unlike pregnancy detection (where the target biomarker hCG is optically invisible), preeclampsia screening targets proteinuria - which is visually detectable via turbidity, foam patterns, and light scattering. Literature shows:

- KIM-1 and MCP-1 urinary biomarkers correlate with early kidney damage in preeclampsia
- Proteinuria causes measurable turbidity changes (AUC=0.984 for turbid vs. clear using CIE Lab* L* threshold)
- Smartphone-based dipstick reading systems achieve >0.95 AUC for UTI detection

Recommendation: If Phase 2 falsifies the pregnancy detection hypothesis, the same imaging and ML infrastructure could be redirected to preeclampsia risk monitoring, where the optical signal is far stronger.

7. Feature Engineering Priorities

Top 5 feature engineering moves, ranked by expected impact and confidence (consensus across reviewers):

Priority	Move	Expected Delta-AUC	Confidence	Key Risk
1	Liquid ROI segmentation + white-balance normalization	+0.03 to +0.06	Medium	Segmentation failures; global normalization leakage
2	Cross-view dispersion features (pairwise Delta-E2000 between photos)	+0.02 to +0.04	Medium	May capture camera physics, not biology
3	Frozen DINOv2 embeddings (ViT-B/14 CLS token)	+0.02 to +0.05	Medium-Low	p >> n; domain mismatch from ImageNet pretraining
4	Nuisance-residualized features (regress out phone, lighting)	+0.00 to +0.03	Medium	May remove real signal if confounded
5	Frozen VideoMAE temporal embeddings	+0.01 to +0.03	Low	Only 68 clips; missing-not-at-random

Critical note: All feature engineering gains are speculative at N=101. No reviewer expects feature engineering alone to push AUC past 0.80 without more data and protocol improvements.

7.1 Current Top Features (V6)

Rank	Feature	Importance	Type
1	edge_intensity_max	0.095	Photo texture
2	t_r_ratio_complexity	0.090	Video temporal
3	turbidity_mean	0.057	Photo clarity
4	color_variance_min	0.049	Photo color
5	t_layla_r_autocorr	0.048	Video spectral
6	ix_r_mean_std_x_b_mean_std	0.048	Interaction (R*B variability)
7	edge_intensity_std	0.044	Photo texture variability

8	layla_r_ratio_std_min	0.031	Photo spectral
9	t_hue_mean	0.029	Video color
10	b_mean_min	0.027	Photo blue channel

8. Regulatory & Clinical Positioning

8.1 Regulatory Classification

Based on the ChatGPT (GPT-5) regulatory analysis and FDA/IMDRF frameworks:

- **SaMD Classification:** Software as a Medical Device. The system analyzes biological specimens to inform health decisions, meeting the IMDRF SaMD definition regardless of hardware independence.
- **FDA Pathway:** De Novo classification is most likely, as there is no clear predicate device for 'pregnancy detection from raw urine images without chemical reagents.' 510(k) would require substantial equivalence to an existing device.
- **Risk Class:** Class II minimum. The system influences reproductive health decisions where false negatives carry significant clinical risk (delayed prenatal care).
- **General Wellness Exemption:** NOT available. A system that outputs 'pregnancy probability' is not a wellness device.
- **Precedent:** FDA issued a warning letter to Biosense (uChek) in 2013, establishing that smartphone-based urine analysis requires premarket clearance.

8.2 Acceptable Use Cases (Current Performance)

Use Case	Feasibility	Rationale
Pre-screening (before buying a pregnancy test)	Possible with caveats	Must never be framed as 'ruling out' pregnancy
Low-resource triage	Possible	Where standard tests are unavailable; better than nothing
Digital triage tool	Possible	Prompts users to seek testing, never replaces it
Wellness consumer app	DANGEROUS	Users will interpret probability as medical diagnosis
Medical device (standalone)	NOT FEASIBLE	Performance far below regulatory requirements
Fertility support (PCOS, IVF monitoring)	Confounded	Supplement use in TTC population makes model unreliable

8.3 Clinical Risk Assessment

The primary clinical risks at current performance levels:

- **False negative ('not pregnant' when pregnant):** Most dangerous. Sensitivity of 60-69% means 31-40% of pregnant women would be missed. This could delay prenatal care. At current performance, the system MUST NOT be positioned as a way to rule out pregnancy.

- **False positive ('pregnant' when not):** Less dangerous but causes unnecessary anxiety, medical visits, and potential financial costs.
- **Over-reliance:** Users may skip confirmatory testing based on model output, especially if probability is presented as high confidence.

9. Phase 2 Protocol Requirements

Based on the synthesized recommendations from all four clinical reviews, the minimum requirements for Phase 2 validation are:

9.1 Study Design

- **Type:** Prospective, blinded, matched case-control study
- **Sample size:** Minimum 300 subjects (150 pregnant, 150 non-pregnant). Ideal: 500-600 for stable calibration plots. External validation requires 200+ events and 200+ non-events.
- **Ground truth:** Serum beta-hCG (quantitative) on the same day as urine imaging. For pregnant subjects: confirmatory ultrasound when clinically indicated.
- **Gestational window:** 4-8 weeks (the relevant window for home pregnancy test replacement)
- **Recruitment sources:** OB/GYN clinics, fertility clinics, and community recruitment. Critical: must include non-pregnant women taking prenatal supplements (TTC population) to break the riboflavin confound.

9.2 Imaging Protocol

- **Lightbox:** Fixed LED lightbox with standardized white illumination (5500K color temperature)
- **Color card:** X-Rite ColorChecker or equivalent in every frame for post-hoc calibration
- **Cup:** Standardized transparent cup, same material and dimensions for all subjects
- **Distance/angle:** Fixed phone mount at 30 cm, perpendicular to liquid surface
- **Captures:** 3 still photos + 1 ten-second video per subject, using standardized study phone AND participant's own phone
- **Metadata:** Log phone model, OS version, capture timestamp, lightbox confirmation

9.3 Matching Variables

Each pregnant subject matched 1:1 with a non-pregnant control on:

- First-morning urine (yes/no)
- Specific gravity or osmolality bin
- Collection time window (within 1 hour)
- Phone model (same device family)
- Folic acid / prenatal vitamin use (CRITICAL)
- Caffeine in previous 6 hours
- Age band

9.4 Pre-Registered Analysis Plan

- **Pipeline:** Frozen before data collection. Single primary model + one pre-specified challenger.

- **Primary metric:** Subject-level AUROC on chronological holdout (last 25% of recruitment)
- **Secondary metrics:** Calibration slope, calibration-in-the-large, Brier score, sensitivity/specificity at pre-locked threshold
- **Statistical test:** DeLong's test for AUROC > 0.60 (one-sided), permutation test at stratum level
- **Subgroup analyses:** Gestational age bands, first-morning vs. later, phone family, supplement use, UTI status, age bands

9.5 Success and Failure Criteria

Criterion	Threshold	Action
External AUROC	>=0.82 (lower 95% CI >=0.75)	Proceed to expanded validation
External AUROC	0.70-0.82	Signal exists but insufficient for standalone use; consider adjunct role
External AUROC	<0.70 or lower CI includes 0.60	TERMINATE pregnancy detection application
Metadata-only baseline	>=0.60	Signal is confounded; do not proceed
Calibration	PPV/NPV aligned with prevalence at pre-set threshold	Required for any deployment
Phone robustness	AUC stable (+/-0.05) across phone families	Required; else restrict to specific devices

9.6 Timeline

Estimated 8-10 weeks end-to-end:

- Weeks 1-2: Protocol finalization, IRB/ethics approval, equipment procurement
- Weeks 3-7: Subject recruitment and data collection (target 15-20 subjects/day)
- Week 8: Data QA, processing, feature extraction
- Weeks 9-10: Locked pipeline evaluation, analysis, reporting

10. Validation Requirements for External Claims

10.1 Minimum External Validation

- **Cohort 1 (Controlled):** ~400 subjects from different recruitment source than development, under standardized rig, with serum ground truth. This tests whether the signal exists.
- **Cohort 2 (Real-world):** ~200 subjects in at-home conditions using their own phones. This tests whether the product works.
- **Model freeze:** All parameters, features, preprocessing, and thresholds frozen before external evaluation.
- **Blinding:** Lab staff blinded to model outputs; model blinded to ground truth.

10.2 Phone Robustness

- Minimum 3-5 phone families (top iOS and Android devices by market share)
- Balanced between classes within each phone family
- Report AUC, sensitivity, specificity, and calibration separately per phone
- Leave-one-phone-family-out development as strongest robustness test

10.3 Calibration

- Calibration-in-the-large, calibration slope, Brier score, reliability diagrams
- PPV/NPV at target prevalence (not case-control prevalence)
- Platt scaling or isotonic regression on separate calibration set if needed
- Post-recalibration results must be clearly separated from pre-calibration

10.4 Required Subgroup Reporting

- Gestational age: <6 weeks, 6-8 weeks, >8 weeks
- Hydration: first-morning vs. later, specific gravity bins
- Supplement use: folic acid/prenatal vitamins yes vs. no
- Phone model: per device family
- Health conditions: UTI, diabetes, kidney disease
- Age: <25, 25-34, >=35
- Protocol adherence: compliant vs. non-compliant captures

11. Honest Assessment & Recommendations

11.1 Where We Stand

With 101 subjects and pure urine visual features:

- **AUC = 0.749 (production) / 0.770 (fusion)** means the model performs better than random (0.50) but far below diagnostic quality (>0.95).
- **The 95% confidence interval** is approximately [0.65, 0.85] given fold variance. We cannot confidently claim performance above 0.65.
- **This is a signal worth investigating** but cannot be used for any clinical, consumer, or reproductive health decision.
- **The path forward is more data and better protocol,** not more sophisticated machine learning.

11.2 What Must Happen Next

In priority order:

6. **1. Riboflavin ablation experiment:** Recruit 30-50 non-pregnant women taking prenatal vitamins. If model accuracy drops to chance, the current signal is behavioral, not biological.
7. **2. Standardized imaging protocol:** Build or acquire a simple lightbox with color card. This alone may improve repeatability by 0.03-0.06 AUC.
8. **3. Phase 2 prospective study:** 300+ subjects with serum beta-hCG ground truth, matched on supplements, with pre-registered frozen pipeline.
9. **4. Resolve N57:** Currently excluded. Final decision needed: reclassify to positive, permanently exclude, or keep as negative.
10. **5. Consider preeclampsia pivot:** If pregnancy detection proves infeasible, redirect to proteinuria monitoring where optical signals are stronger.

11.3 What We Tell Doctors

We have built a research prototype that detects *something* in urine images that correlates with pregnancy status at a level better than chance. We do not yet know whether that something is a genuine physiological signal, a prenatal vitamin artifact, or an acquisition confound. We are committed to finding out through a rigorous, pre-registered study with medical-grade ground truth. We are not claiming this replaces pregnancy tests. We are asking: is there a signal worth pursuing?

Research Use Only. Experimental probability model. Not a diagnostic test.

This document is intended for clinical research evaluation only.

12. References & Literature

12.1 Primary Literature

- **Bustam et al. (2023).** 'Accuracy of smartphone camera urine photo colorimetry as indicators of dehydration.' Digital Health. PMC10474791
- **Noor Azhar et al. (2023).** 'Improving the reliability of smartphone-based urine colorimetry using a colour card calibration method.' JMIR AI.
- **Flaucher et al. (2022).** 'Smartphone-Based Colorimetric Analysis of Urine Test Strips for At-Home Prenatal Care.' IEEE J Transl Eng Health Med.
- **Shen et al. (2025).** 'Longitudinal urine metabolic profiling and gestational age prediction in human pregnancy.' Briefings in Bioinformatics.
- **Zhang et al. (2023).** 'Development of a Urine Metabolomics Biomarker-Based Prediction Model for Preeclampsia during Early Pregnancy.' Metabolites.
- **Prabhakar et al. (2022).** 'Exploration of urinary metabolite dynamicity for early detection of pregnancy in Murrah buffaloes.' Scientific Reports.
- **Ra et al. (2017).** 'Smartphone-Based Point-of-Care Urinalysis Under Variable Illumination.' IEEE J Transl Eng Health Med. PMC5764119
- **Pohanka (2024).** 'Urine Test Strip Quantitative Assay with a Smartphone Camera.' Int J Anal Chem. PMC10963100
- **Kim (2024).** 'AI in Diagnostics: Enhancing Urine Test Accuracy Using a Mobile Phone-Based Reading System.' Ann Lab Med. PMC11788702

12.2 Clinical Guidelines

- NICE NG126: Antenatal care for uncomplicated pregnancies
- ACOG Committee Opinion: Methods for Estimating the Due Date (2017)
- FDA: Home Use Tests - Pregnancy (regulatory guidance)
- FDA: In Vitro Diagnostics (device classification)
- FDA: De Novo Classification Request (submission guidance)
- IMDRF SaMD: Clinical Evaluation (IMDRF-TECH-170921-SAMD-N41)
- EU MDR 2017/745 and IVDR 2017/746: Software qualification guidance
- EFLM Urinalysis Guideline (2024)

12.3 Independent Clinical Reviews

- Perplexity Clinical Review (May 4, 2026) - Q1-Q5 structured technical assessment
- Gemini Clinical Review (May 4, 2026) - Comprehensive Hebrew literature review on prenatal diagnostics
- ChatGPT/GPT-5 Clinical Review (May 4, 2026) - Q1-Q5 structured assessment with regulatory analysis
- ChatGPT/GPT-5 Hebrew Clinical Review PDF (May 2026) - 7-page referenced clinical review